

Lab 5: Introduction to pavo — Quantifying Reef Fish Color Patterns

Week 6

EEB 187 — Ecology & Evolution of Color

Overview

This is the first computational analysis lab of the course. You will use the R package **pavo** to extract color pattern metrics from reef fish photographs.

Format: Instructor-led demo on three reef fish from three different families + independent practice on three fish from your group's assigned lineage.

- **Demo + walkthrough** (instructor + Rosamari): 35–40 min
- **Independent practice** (3 fish from your lineage): 30–35 min
- **Wrap-up + cross-lineage comparison:** 5–10 min

Bring:

- Laptop with R, RStudio, and pavo installed (see `pre-lab-installation.qmd`)
- The lab bundle: `labs/demo-pavo-week6/` (download from BruinLearn)
- Three image files from your assigned lineage saved locally before lab — see Part B Step 1

Part A: The demo — three reef fish, three pattern axes

We're starting with three fish from three different families. Each fish has a fundamentally different pattern type. The pedagogical claim of the lab: pavo's adjacency framework can distinguish these patterns in numbers — specifically with the **A statistic** (aspect ratio).

Demo fish	Family	Pattern	Predicted A
Sergeant major (<i>Abudefduf saxatilis</i>)	Pomacentridae	Vertical bars	$A > 1$
Cleaner wrasse (<i>Labroides dimidiatus</i>)	Labridae	Horizontal stripe	$A < 1$
Mandarin fish (<i>Synchiropus splendidus</i>)	Callionymidae	Reticulate maze	$A \approx 1$

Follow along with `scripts/walkthrough.R`. Rosamari is presenting the slide deck; the R script is on your laptop.

Record the demo-fish metrics here for reference later:

Metric	Sergeant major	Cleaner wrasse	Mandarin fish
k			
m			
A			
Jc			
Jt			
m_dS			
m_dL			

Part B: Your turn — three fish from your assigned lineage

Your group has been assigned a reef fish lineage (typically a family). Today's job: pick **three species from your lineage** that show **three visually different pattern types**. Run all three through the pavo pipeline. Compare your results to the demo fish from Part A.

Step 1: Pick three fish

i Where to find images depending on your launch path

Zero-hassle option (works in every launch path including Binder/Codespaces): the repo ships with 5 species per assigned lineage in `images/student-fish/<your-lineage>/`. Pick 3 of those and you're done — no uploading, no internet hunting. If you want to use a fish that isn't pre-loaded:

- **Native install / Docker:** save the JPG anywhere on your laptop and pass the full path to `segment_fish()` (e.g., `~/Downloads/my_fish.jpg`). Your work persists between RStudio sessions.
- **Binder (cloud, ephemeral):** in RStudio's *Files* pane (bottom-right by default), click the **Upload** button → pick your JPG → upload it into the project root or `images/student-fish/<your-lineage>/`. Then refer to it as `"images/student-fish/<your-lineage>/my_fish.jpg"`. **Important — Binder sessions expire after 90 minutes of inactivity and uploaded files are NOT saved.** Finish your worksheet in one sitting, or download the segmented JPG (Files pane → check the box → *More* → *Export*) before logging off.
- **Codespaces (cloud, persistent):** drag-drop the JPG into the VS Code file explorer, then refer to it by relative path. Files persist between visits, so you can come back later.

Browse the available images for your assigned lineage. (FishView, FishBase, your group's curated set, or `images/student-fish/<your-lineage>/` which is pre-loaded for every launch path.) Pick three species that look as different as you can find — for example:

- One with strong **vertical bars** (you predict $A > 1$)
- One with a **horizontal element** like a stripe or band (you predict $A < 1$)
- One that's **spotty, reticulate, or solid** (you predict $A \approx 1$)

If your lineage doesn't have all three pattern types, pick the three that span the widest pattern range available.

Write your three species here:

Lineage: _____

Fish 1 (predict $A > 1$): _____

Fish 2 (predict $A < 1$): _____

Fish 3 (predict $A \approx 1$): _____

Step 1.5: SEGMENT — remove the background from each image

This is the most important pre-processing step. Without it, pavo's `k-means classify()` runs over the WHOLE image including water, reef, and tank glass. Background pixels become “color classes,” and the resulting metrics describe the photograph rather than the fish.

The 3 demo fish you saw in Part A were segmented using **rembg**, an AI background-removal tool. The demo's dramatic A values (1.54 / 0.60 / 1.02) depend on this segmentation step. You have two options:

Path 1 (recommended): rembg via `segment_fish()` R helper

The `segment_fish()` function in `walkthrough.R` wraps `rembg` + `ImageMagick` into one R call. **One-time setup** (you should have done this in pre-lab Step 6):

```
# Install uv (handles rembg's Python + dependencies automatically):
brew install uv                # macOS
curl -LsSf https://astral.sh/uv/install.sh | sh    # Linux
# Plus ImageMagick:
brew install imagemagick       # macOS
sudo apt install imagemagick   # Linux
```

Per-fish use (in R during lab):

```
seg_path <- segment_fish("path/to/raw_fish.jpg")
my_fish  <- getimg(seg_path)
```

The result is a JPG with the fish on a clean white background, saved as `<name>-segmented.jpg`. If `segment_fish()` errors out, fall back to Path 2.

 If `segment_fish()` fails with “rembg failed (or not installed)”

R may not see your shell's PATH. First confirm the tools work in your terminal:

```
uvx --from "rembg[cpu,cli]" rembg --help
magick -version
```

If both work in terminal but `segment_fish()` still fails, R's PATH is incomplete. Check it:

```
Sys.getenv("PATH")
```

If `/opt/homebrew/bin` (Homebrew on macOS) or `~/.local/bin` isn't there, add it to `~/.Renviron`:

```
PATH=/opt/homebrew/bin:/usr/local/bin:/usr/bin:/bin:~/.local/bin
```

...and restart RStudio. If you're still stuck, use Path 2 (web tool) or Path 3 (manual crop) below — both work without needing R to find any shell tools.

Path 2 (no install, web-based): segment with remove.bg or photoroom.com

Drag-drop your fish JPG into one of these free tools:

- [remove.bg](#) — fastest, no signup
- [photoroom.com](#) — also free
- [clipdrop.co/remove-background](#) — Stability-AI tool

Each returns a transparent PNG. Save the PNG into your `images/student-fish/<your-lineage>/` folder. Then in R, use the `composite_to_white()` helper to add a white background and standardize the canvas:

```
source("scripts/helpers.R") # loads composite_to_white()
seg_path <- composite_to_white("images/student-fish/<lineage>/your-fish.png")
my_fish <- getimg(seg_path)
```

This gives you the same `*-segmented.jpg` you'd get from `segment_fish()` — pavo's classify-with-bkg path will work exactly the same way.

Path 3 (fallback, no install, no internet): manual crop

Open each fish image in **Preview.app** (or any image editor), drag a selection rectangle as tight to the fish as possible without cutting any fins, then Tools → Crop and save. Load with `getimg("path/to/cropped.jpg")` — skip both `segment_fish()` and `composite_to_white()`.

💡 Tradeoff for the manual-crop path (Path 3)

Cropped images still have water/reef in the corners. **Your A values will be less dramatic than the demo** (closer to $A=1$ for all three) because some background pixels will pollute the k-means classification and the adjacency grid. The directional ordering still holds — vertical-bar fish > 1 , horizontal-stripe fish < 1 — just with smaller effect sizes. Note in your worksheet which path you used. The rubric accommodates all three. Step 2 below has a different code block for the manual-crop path: drop the `bkgID = my_bkg`, `exclude = "background"` arguments since there's no white background class to exclude. (Paths 1 and 2 both produce a white-background JPG and use the standard code block.)

Comparing the three folders for the demo fish

Look at `images/demo/raw/` (uncropped originals) and `images/demo/segmented/` (rembg output) to see exactly what segmentation does. Run `getimg()` on both and compare the resulting `classRGB` palettes — the raw ones include a “blue water” class; the segmented ones don’t.

Step 1.6: REGISTER each image to left-lateral view

Standard ichthyology convention: head on the LEFT, tail on the RIGHT. Before classifying, look at each of your three images and confirm the head is on the left. If a fish is facing the wrong way, flip it horizontally:

```
# Defined in walkthrough.R
flip_to_left_lateral("path/to/my_fish.jpg")
```

Or from a terminal (macOS / Linux with ImageMagick installed):

```
magick my_fish.jpg -flop my_fish.jpg
```

This step matters because:

- Predictions like “vertical bars” presume the canonical orientation
- Cross-fish comparison in the next pavo session expects all images registered the same way
- Same goes for any biology paper you’ll publish with this kind of analysis

Note: horizontal mirroring does NOT change the A, m, Jc, or other adjacency metrics — they’re invariant to left/right reflection. Registration is for visual consistency and comparative validity, not metric correctness.

Step 2: Run the pipeline on each fish

For each of your three segmented + registered fish, run:

```
my_fish <- getimg(seg_path)          # path returned by segment_fish() or your crop
plot(my_fish)

# k = (number of distinct fish colors) + 1 if you used rembg (the +1 is for
# the white background). For manual-crop path, just the fish-color count.
my_k    <- 4
my_class <- classify(my_fish, kcols = my_k)
plot(my_class)
```

```

# Identify and exclude white background (only for rembg path)
my_bkg <- pick_white_bg(my_class) # = which.max(rowSums(classRGB))
my_cd <- simple_coldists(my_class)
my_adj <- adjacent(my_class, xpts = 200, xscale = 5, coldists = my_cd,
                  bkgID = my_bkg, exclude = "background")
# For the manual-crop path: drop the bkgID + exclude args.

# Pull the metrics
my_metrics <- tibble(
  k    = pick_col(my_adj, c("k", "kcols")),
  m    = pick_col(my_adj, c("m", "m_r")),
  A    = pick_col(my_adj, c("A", "Asp")),
  Jc   = pick_col(my_adj, c("Jc")),
  Jt   = pick_col(my_adj, c("Jt")),
  m_dS = pick_col(my_adj, c("m_dS", "Cs_dS")),
  m_dL = pick_col(my_adj, c("m_dL", "Cs_dL")),
)

```

! Load the helpers FIRST

All the helper functions (`pick_col()`, `simple_coldists()`, `pick_white_bg()`, `segment_fish()`, `composite_to_white()`, `flip_to_left_lateral()`) live in `scripts/helpers.R`. Run this **once at the top of your session** before anything else:

```
source("scripts/helpers.R")
```

If you skip this and try to call e.g. `simple_coldists(my_class)`, you'll get **Error: could not find function "simple_coldists"**. Source it once and every helper is in scope for the rest of your session.

Step 3: Record your results

Metric	Fish 1	Fish 2	Fish 3
Species			
k			
m			
A			
Jc			
Jt			

Metric	Fish 1	Fish 2	Fish 3
m_dS			
m_dL			

Step 4: Compare to the demo fish

Look at your three fish's A values vs. the three demo fish (Sergeant major, Cleaner wrasse, Mandarin) from Part A.

(a) Did the A statistic correctly classify your three fish by pattern orientation?

____ YES — my barred fish has $A > 1$, my horizontal fish has $A < 1$, my reticulate/spotty fish has $A \approx 1$

____ PARTIALLY — explain which one(s) didn't behave as expected:

____ NO — none of them landed where I predicted. Speculate why:

Notes:

(b) Cross-lineage comparison. Pick ONE of your three fish whose pattern most resembles one of the demo fish. Compare their full metric profiles (m, A, Jc, m_dL).

My fish: _____

Most similar demo fish: _____

What's similar in the metrics:

What's different in the metrics:

Step 5: Biological interpretation

In 4–6 sentences, answer:

(a) **Within your lineage:** are the three fish you picked more similar to each other in metric space than they are to fish from other families? Or are some of your three fish a closer metric match to the demo fish than to each other? What might that suggest about pattern evolution within vs. across reef fish lineages?

(b) **Selection hypothesis:** for ONE of your three fish, propose a selection pressure that might have favored its pattern. Pick from: crypsis, aposematism, species recognition, sexual signaling, predator deflection (eyespot), disruptive coloration, or “I have a hypothesis but I’m uncertain — here’s why.”

Submitting

Submit by end of day to BruinLearn. Either:

- A rendered HTML or PDF of this .qmd with your tables filled in, OR
- A 1-page Word/Google doc with the **3 metric tables** + the **comparison answers** + the **biological interpretation**.

The R code does not need to be polished. The biological reasoning is the assessable part.

What's next

There will be a **second pavo session** later in the term. We'll build a metric matrix across many species per lineage, cluster fish by pattern, and compare the clusters to phylogeny. Today's lab gives you the tool; next time we use it to ask comparative questions.

Common errors and their fixes

Error in getimg: cannot open the connection — file path is wrong. Use `getwd()` to see where R thinks you are. In RStudio: Session → Set Working Directory → To Source File Location.

Classify image looks weird (washed out, only 2 colors) — try a different `k`. If your fish has 5 visible color regions, `k = 5`.

Error: object 'classify' not found — you didn't run `library(pavo)`. Run it.

Adjacency runs but everything is NA — your image is empty / all-white / all-black. Re-load with `getimg()` and check `plot(img)` actually shows the fish.

The metrics table looks empty in some columns — `pavo` column names vary by version. The `pick_col()` helper in `walkthrough.R` falls back gracefully; if any metric is `NA`, look at `colnames(my_adj)` to find the actual name in your version.

My A values for the 3 fish don't match my predictions — this is real! Common causes: (1) `k` is too small or too large, collapsing or splitting biological color regions; (2) the image background isn't well-segmented (background bleeds into the analysis); (3) the fish actually doesn't have the pattern you thought it had — `pavo` is a more sensitive eye than yours. Note this in your interpretation; it's good science.

What this lab is NOT

We are not (yet) doing:

- Spectral analysis from a spectrometer (different `pavo` functions: `procspec`, `peakshape`, etc.)
- Visual modeling through a fish's eyes (`vismodel`, `coldist`, JNDs)
- Many-species comparative analysis (next `pavo` session)
- Phylogenetic comparative methods on color (Lab 9, end of term)

Today we extract pattern metrics from three reef fish, predict-and-test, and compare across lineages. That alone is a substantive analytical exercise.